

Course “Object-Oriented Programming in C++”**Week №7*****Homework*****Exercise 1.**

There is a string of characters. Semicolon is a sign of its end. Curly brackets, round, square brackets can be in a string. Opening and closing parentheses can be in a string.

It is necessary to check the placement of brackets. It is necessary to follow these rules:

1. Each opening parenthesis must have the same closing parenthesis on the right. The reverse should also be right.
2. Closing and opening brackets of different types must be correctly placed.
 - **Example of correct string:** $(\{x-y-z\}^*[x+2y])-(z+4x))$
 - **Example of incorrect string:** $([x-y-z]^*[x+2y])-\{z+4x\})$.

If all rules are observed, output the information about the correctness of string, otherwise show the string until the first error will emerge.

Exercise 2.

Create a simulation model "minibuses stop". You must enter the following information: the average time between occurrences of passengers at the minibus stop at different times of the day, the average time between the appearances of minibuses at the bus stop at different times of day, type of stop (final or not).

It is necessary to determine: the average time of stay of the person at the bus stop, sufficient time interval between arrivals of minibuses in order not more than N people simultaneously were at the stop. The number of vacant seats in the minibus is a random variable

Exercise 3.

Develop an application that simulates the print queue. Clients should send requests to the printer; each of them has its own priority. Every new customer enters the queue according to its priority. It is necessary to preserve the print statistics (user, time) in a separate queue. Output the statistics to the screen.